SIMATS ENGINEERING

SWETHA.B
192412215

EXPERIMENT 19

ITA0606
MACHINE LEARNING

Implementation of Naive Bayes classification for Bank Loan prediction

Code

```
# Naive Bayes Classification for Bank Loan Prediction

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

# Read dataset
df = pd.read_csv("play_tennis (1).csv")

# Remove missing values
df = df.dropna()

# Separate input features and target
X = df.iloc[:, :-1]
y = df.iloc[:, -1]

# Convert categorical features into numerical values
for column in X.columns:
    if X[column].dtype == 'object':
        X[column] = LabelEncoder().fit_transform(X[column])

# Encode target
encoder = LabelEncoder()
y = encoder.fit_transform(y)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Create Naive Bayes model
model = GaussianNB()

# Train
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy (%):", accuracy * 100)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)
```

```python
ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=encoder.classes_
).plot()

plt.title("Bank Loan Prediction using Naive Bayes")
plt.show()
```

OUTPUT:

```
Accuracy: 0.6666666666666666
Accuracy (%): 66.66666666666666

Confusion Matrix:
[[0 1]
 [0 2]]
```